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*DEPARTMENT OF CIVIL, ENVIRONMENTAL
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The Conference on ***C****omputational* ***M****ethods in* ***W****ater* ***R****esources*
June 28 – July 2, 2026, Bologna, Italy

Session 17: Model coupling, domain decomposition, and solvers for multiphysics problems

GReS: A Novel Multi-Physics and Multi-Domain Prototyping Platform for Coupled Subsurface Problems

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Computational methods are essential tools for assessing subsurface engineering applications: geothermal energy production, underground gas storage, carbon sequestration etc.

These processes are inherently

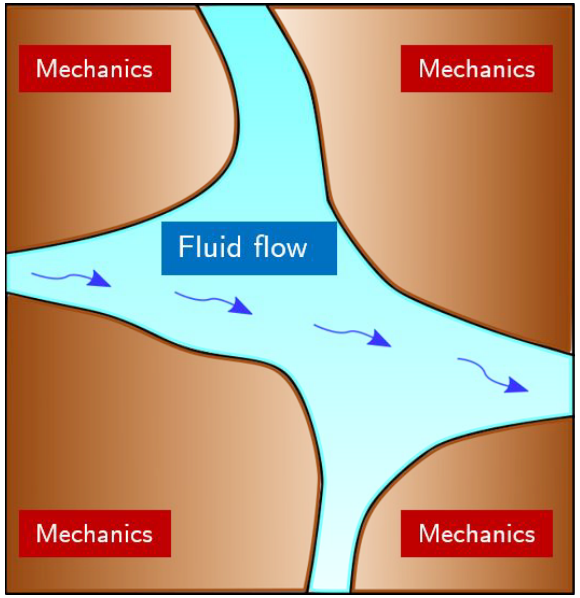
- **MULTI-PHYSICS**

fluid flow, mechanical deformation, fracture propagation, heat transfer, chemical reactions.

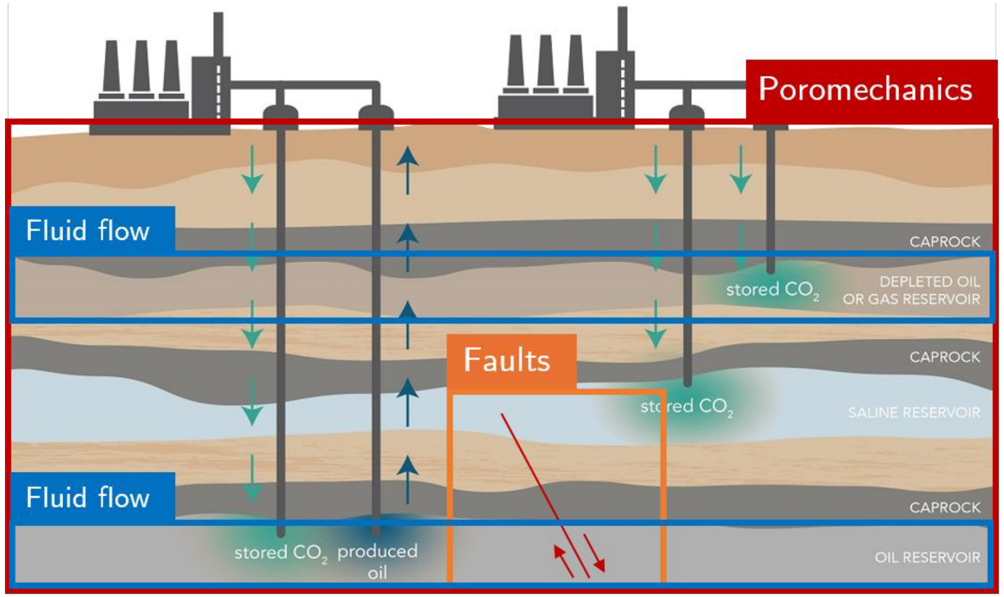
- **MULTI-SCALE**

different phenomena happens at different scales of interest (in space and time).

Pore scale



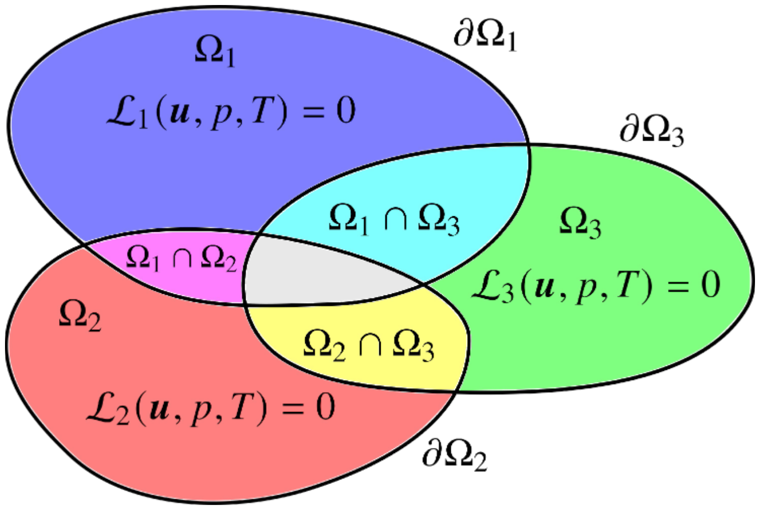
Field scale

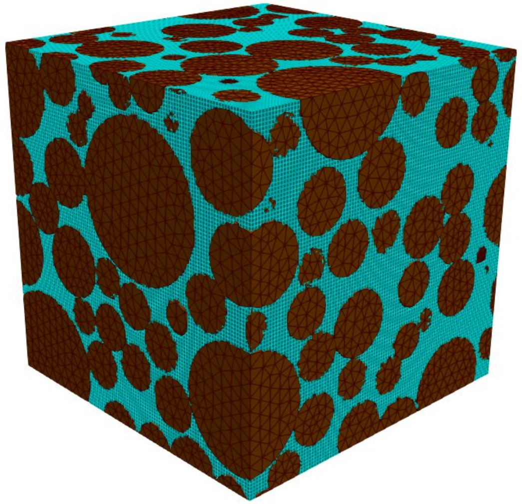


CO2 sequestration – Source: California Air Resources Board

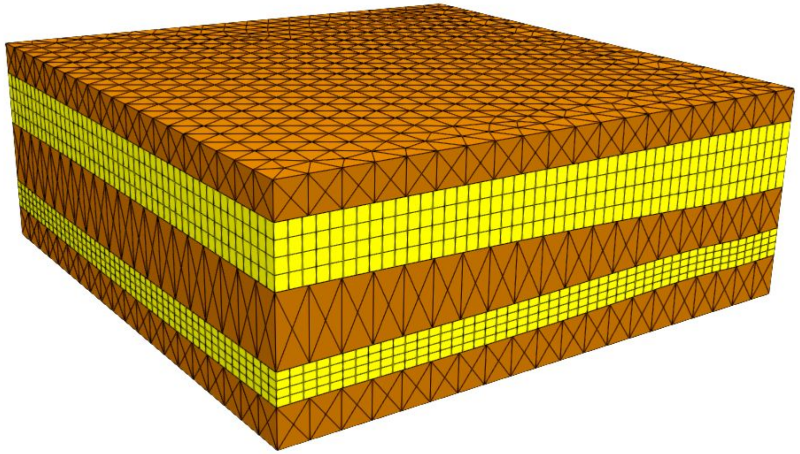
To (partially) tackle the multi-physics and multi-scale nature of the problem, we develop a framework that:

- subdivide the overall domain into subdomain and targets arbitrary combination of physics in each subdomain.
- allow the subdomains to be geometrically independent grids;
- facilitates the use of different discretization techniques in different domains;





Pore scale fluid-structure interaction model



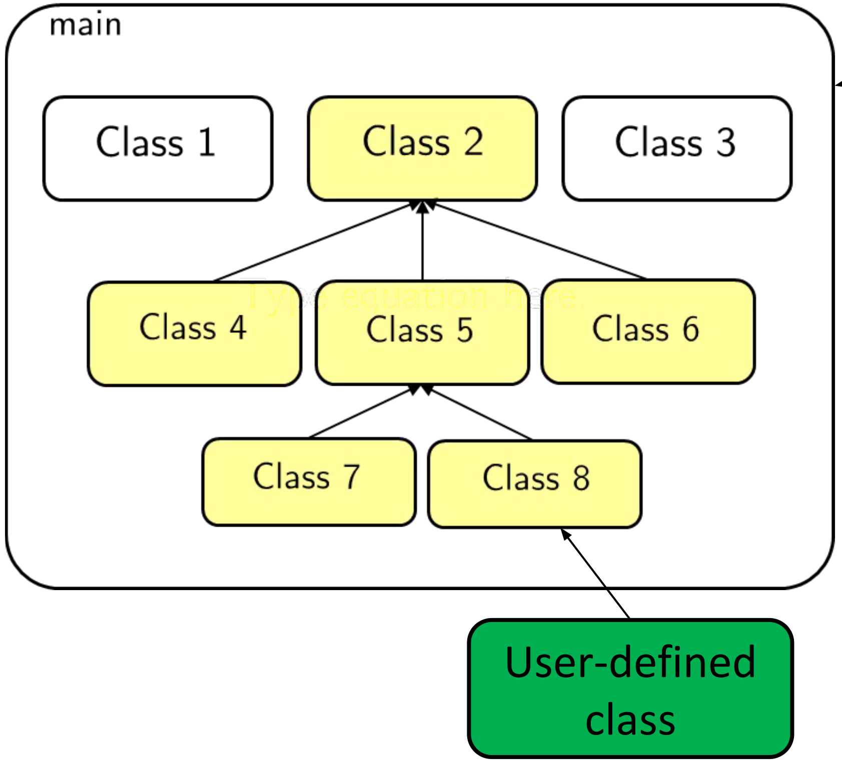
Multi-aquifer non-conforming model

We implement this framework in GReS, an **open-source** computational platform which enables **fast prototyping** of customized **coupled formulations;**

Scan to reach the GitHub repository

Prototyping in MATLAB

GReS is written in MATLAB, allowing:



simplicity – low entry barrier

- High level programming interface
- Ease of use
- Cross-platform compatibility

modularity – needed for multi-physics codes
each physical model is a different MATLAB class

efficiency – not only toy problems

- Powerful vectorization and logical indexing
- Easy interface with compiled languages through MEX-code
- Effective memory management tools

widespread use – it is one of the most used software in academia and industry for numerical models

interface with other software – e.g., MRST

Available features and modules

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Physics

- Small strain solid mechanics;
- Single Phase Flow
- Unsaturated flow
- Biot coupled poromechanics
- Evolving depositional flow
- Lagrange multipliers Frictional contact mechanics
- Embedded fracture mechanics

Discretizations

- Finite Element (1st and 2nd order)
- Finite Volumes (TPFA)
- Mortar finite elements
- Embedded finite elements

Constitutive laws

- Linear elasticity
- Elasto-plasticity (Drucker-Prager)
- Visco-elasticity (SSCM)

Mesh

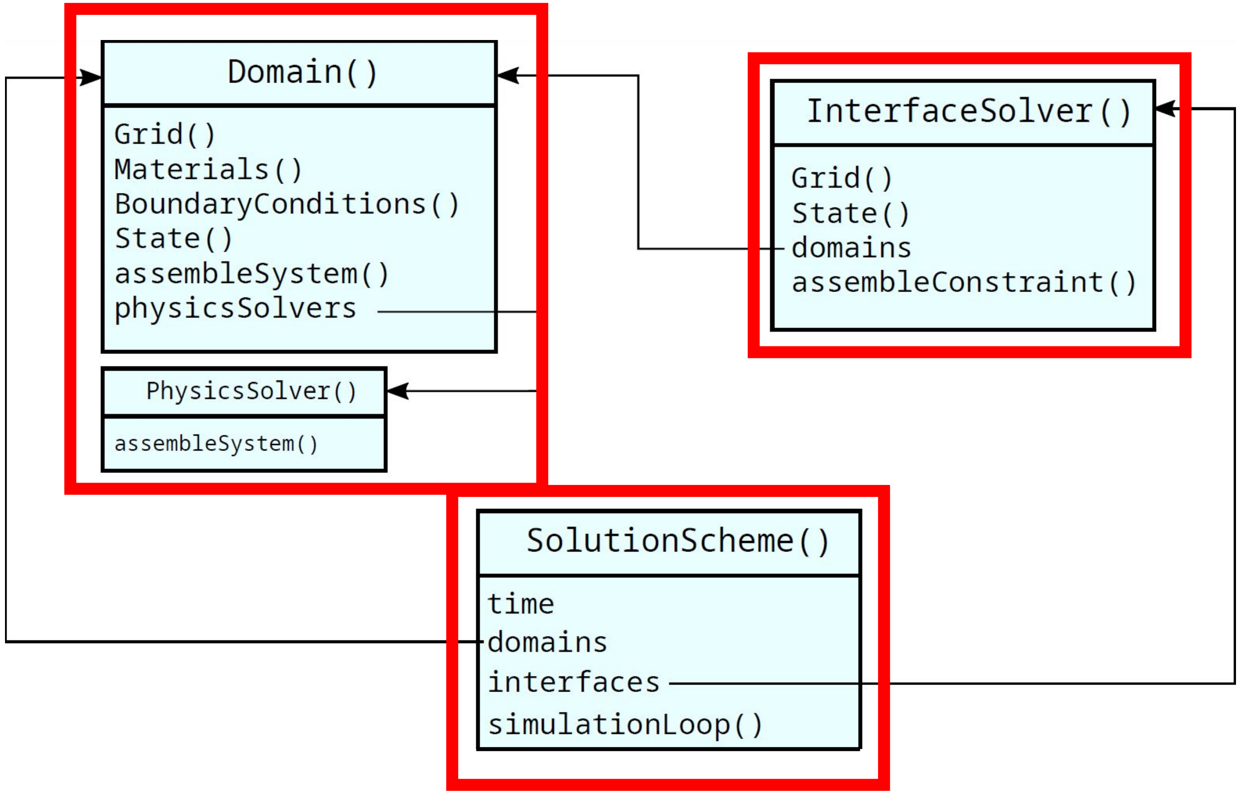
- Tetrahedra
- Hexahedra
- Non-conforming meshes
- VTK input/output

GReS consists of three main classes. The user can derive from this base classes and provide a customized implementation.

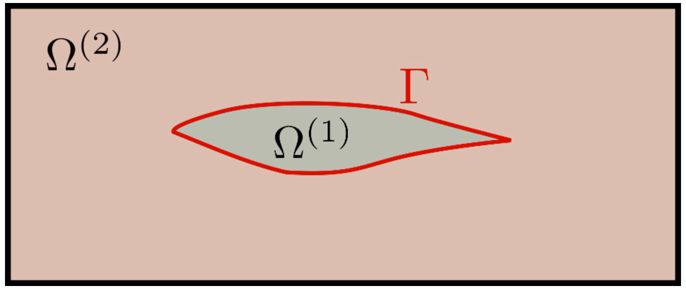
Domain(): container for one or more PhysicsSolver() objects. Each PhysicsSolver() registers a set of variables and implements a set of governing equations.

InterfaceSolver(): the class responsible for coupling one or more variable fields shared by two neighboring domains.

SolutionScheme(): implements the solution strategy used to advance the simulation in time. This strategy may be either general-purpose or tailored to a specific combination of physics solvers.



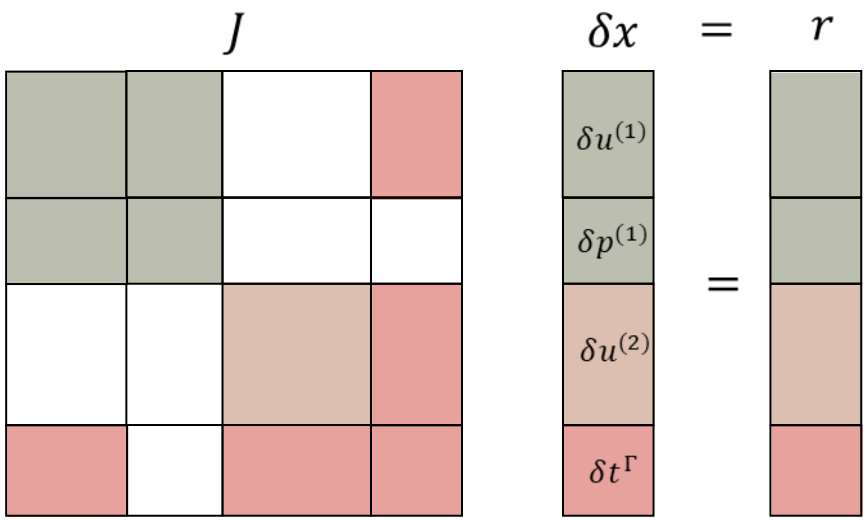
A simulation example



$\Omega^{(1)}$: Coupled pressure $p^{(1)}$ and displacements $u^{(1)}$

$\Omega^{(2)}$: Displacements $u^{(2)}$

Resulting fully coupled linearized system



domains(1) | domains(2) | interface

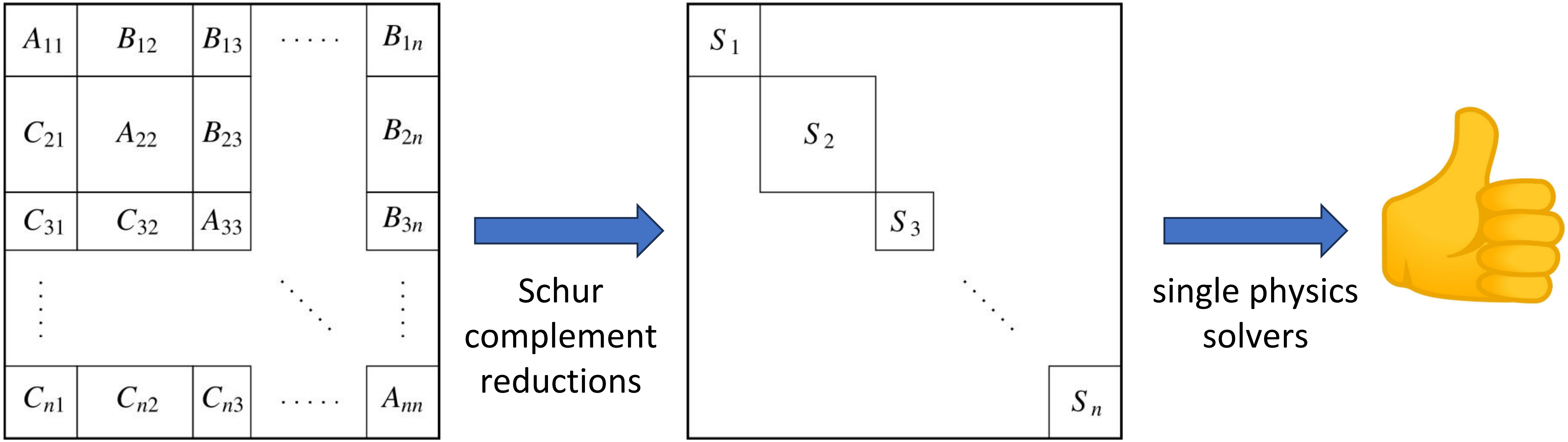
```
% import separate grids from vtk files
grid1 = importMesh('domain1,vtk');
grid2 = importMesh('domain2.vtk');

% define material
mat = Materials();
mat.addSolid('name',"rock","specificWeight",21.0,'cellTags',1);
mat.addConstitutiveLaw("rock",...
  "Elastic",'youngModulus',1.0e6,'poissonRatio',0.25);
mat.addFluid('compressibility',4.59e-7,'dynamicViscosity',1e-3);

% build the Domains
domain1 = Domain('grid',grid1,...
                 'materials',mat,...
                 'boundaries',Boundaries(grid1,'bc1.xml'));
```

```
% add the physics solvers
domains(1).addPhysicsSolver("BiotFullyCoupled");
domains(2).addPhysicsSolver("Poromechanics");

% define the interface
interf = struct('masterDomain',1,'slaveDomain',2);
interface = InterfaceSolver.add("MeshTying",domains,interf);

% set the solution strategy and run the simulation
simparams = SimulationParameters(input.SimulationParameters);
solver = NonLinearImplicit('simulationparameters',simparam,...
                           'domains',domains,...
                           'interfaces',interfaces);
solver.simulationLoop();
```

Solution of the block linear system

To deal with a block linear system, there are two main ingredients:

- Schur complement reductions: strategies to (approximatively) decouple the different physics
- Single physics solvers: solvers for all the decoupled problems

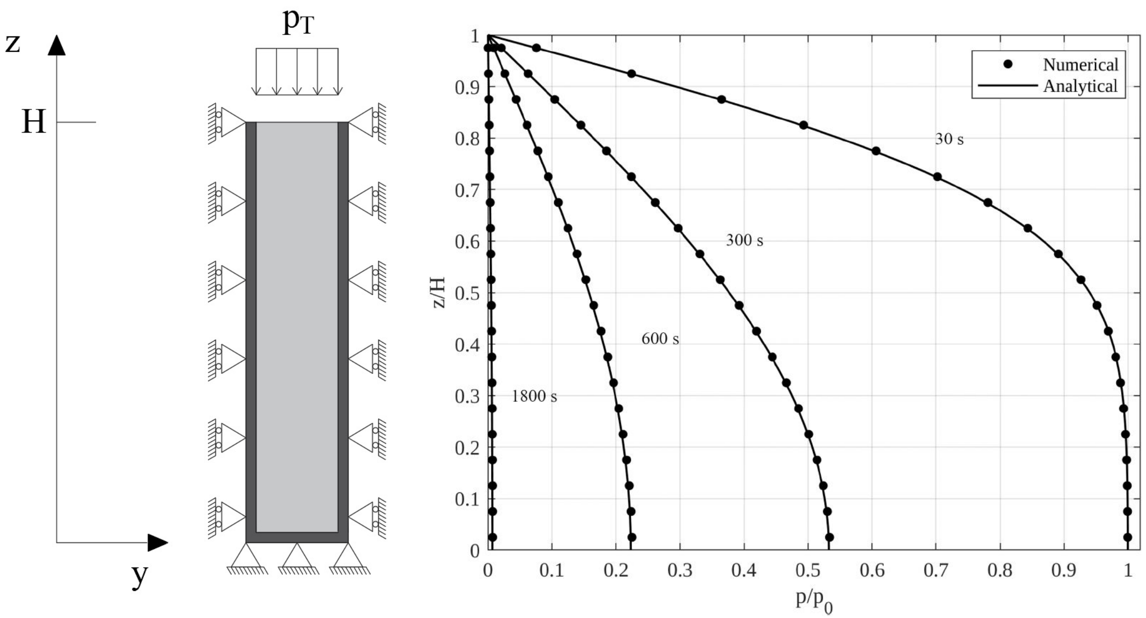


MULTI-GRID

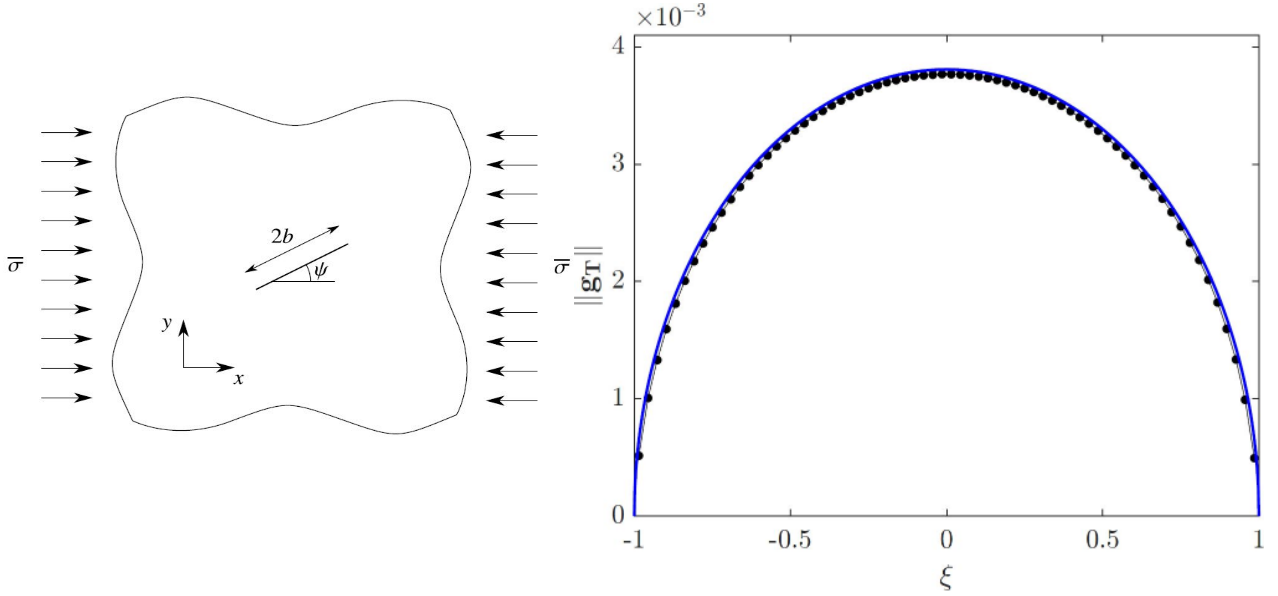
Available benchmarks (Biot problem)

GReS comes with a set of verification benchmarks and use cases.

Terzaghi consolidation
Biot coupled flow and mechanics



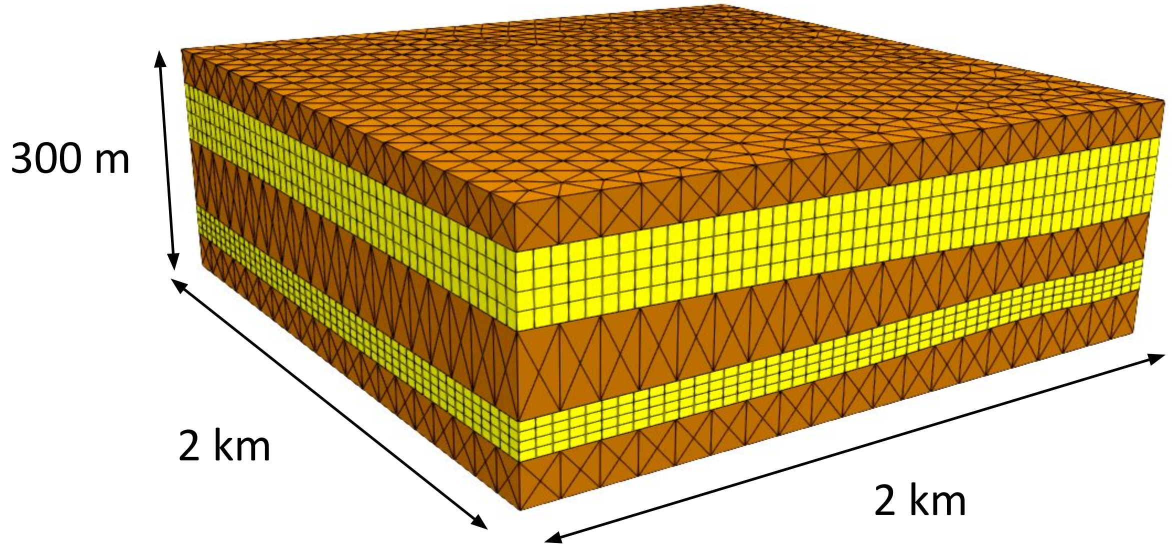
Single fracture under compression
Frictional contact mechanics with Lagrange multipliers



The code is also documented with hands-on tutorials

Test case - Multi-layered aquifer model

The proposed framework is tested against a multi-aquifer model with non-conforming layers.



It is a complete multi-physics multidomain simulation with:

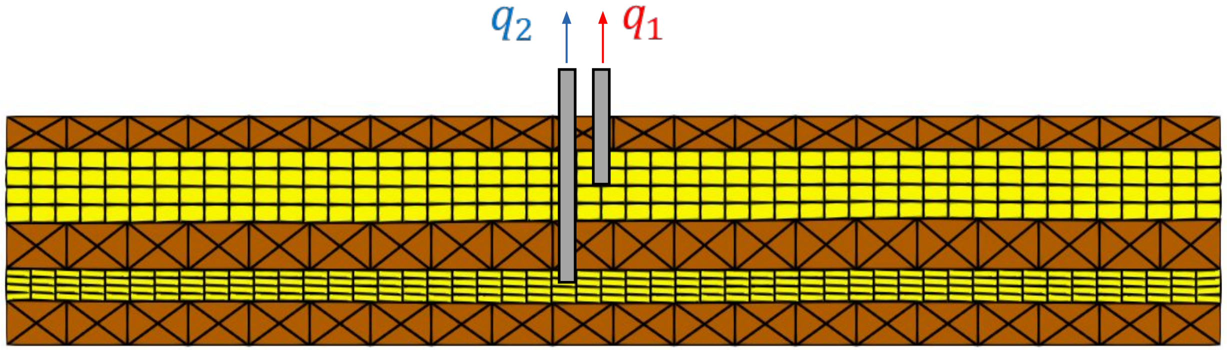
- Non-conforming meshes
- Non-conforming geometries
- Different discretization schemes (FEM in **aquitards** and FEM+FV in aquifers)

Clay layer
Poromechanics (FEM)

$$\nabla \cdot \boldsymbol{\sigma} + \boldsymbol{b} = \mathbf{0}$$

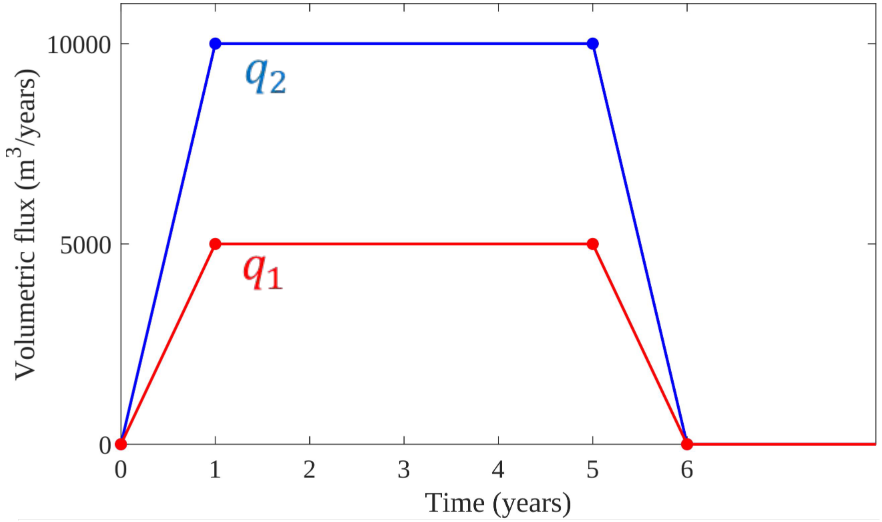
Sandy layer
Biot coupling (FEM+FV)

Multi-layered aquifer model – results I

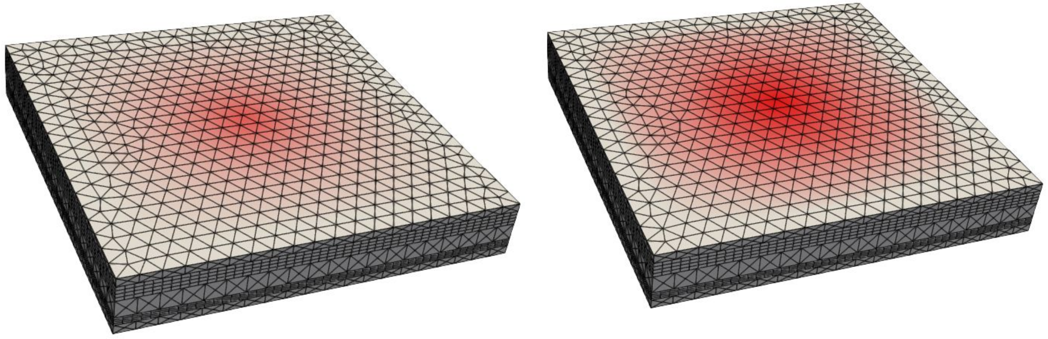


Cells: 28921 Nodes: 28380 DoFs: 104348

Sand layer	
	20000 kPa
	0.3
Biot coefficient	1.0
Clay layer	
	10000 kPa
	0.3



Vertical displacements

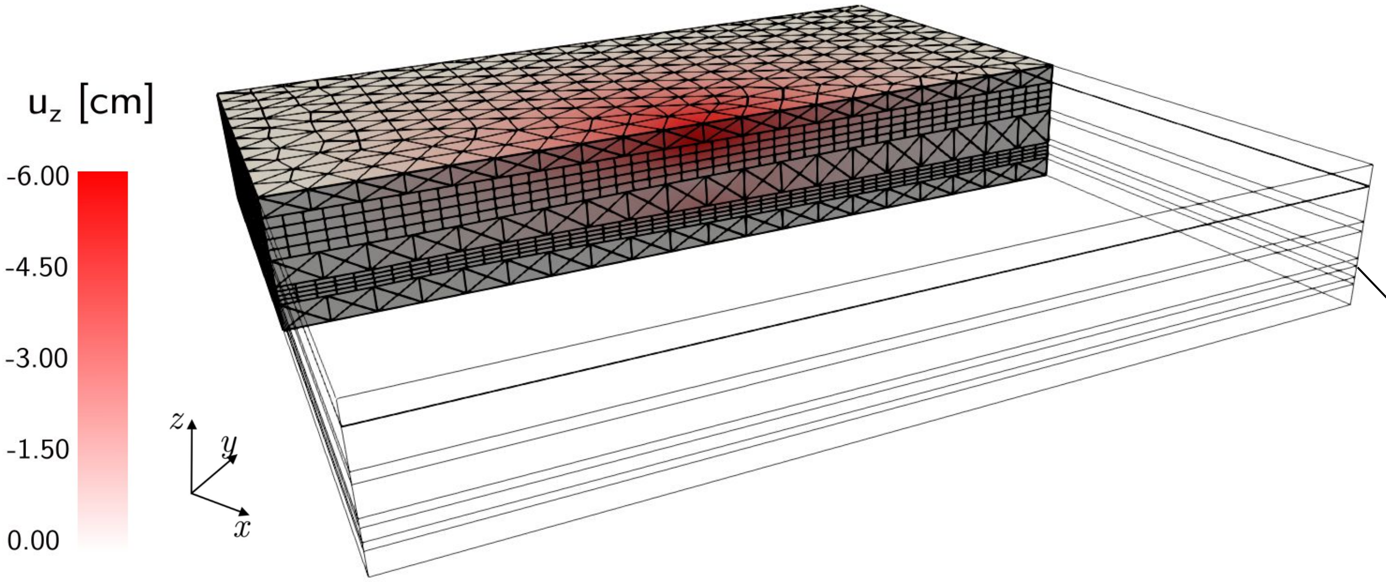


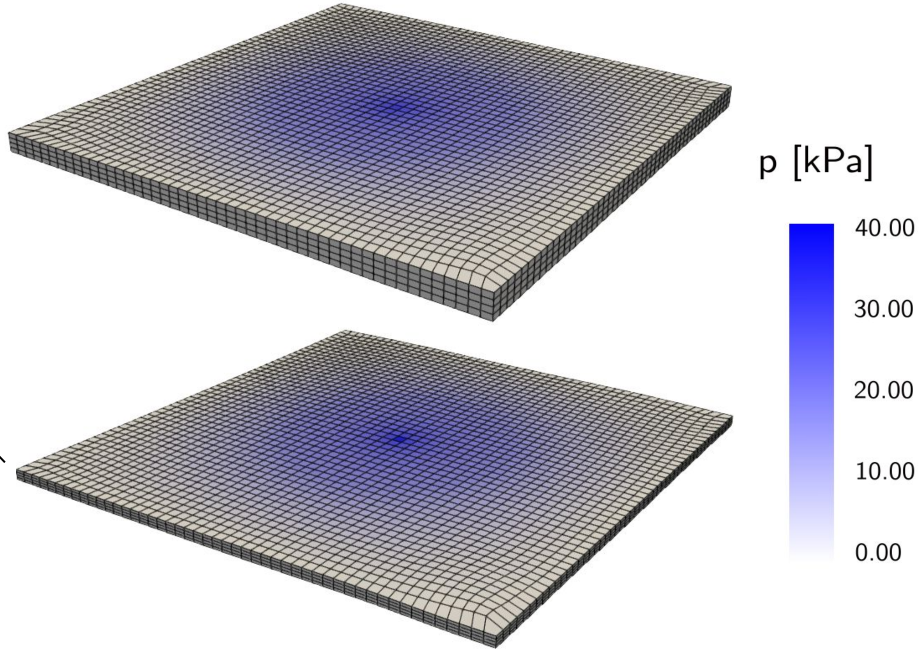
t = 1 year t = 5 years

Multi-layered aquifer model – results II

Vertical displacements (section)

Pressure distribution in aquifer layers



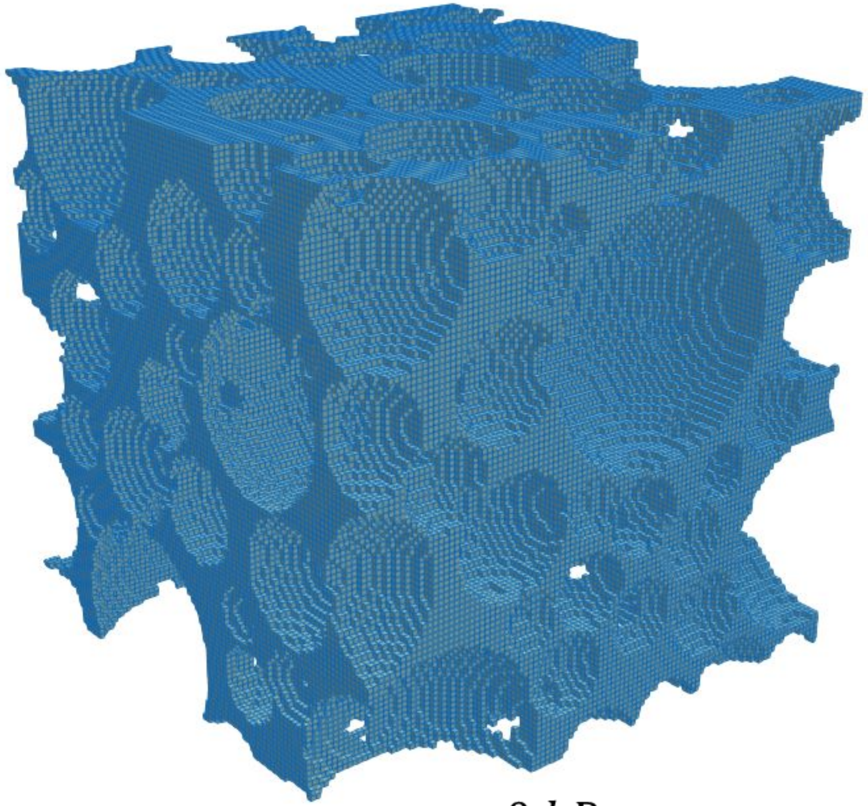


t = 5 years

Results confirm the geometric flexibility of our approach and applicability to coupled fluid-mechanics applications.

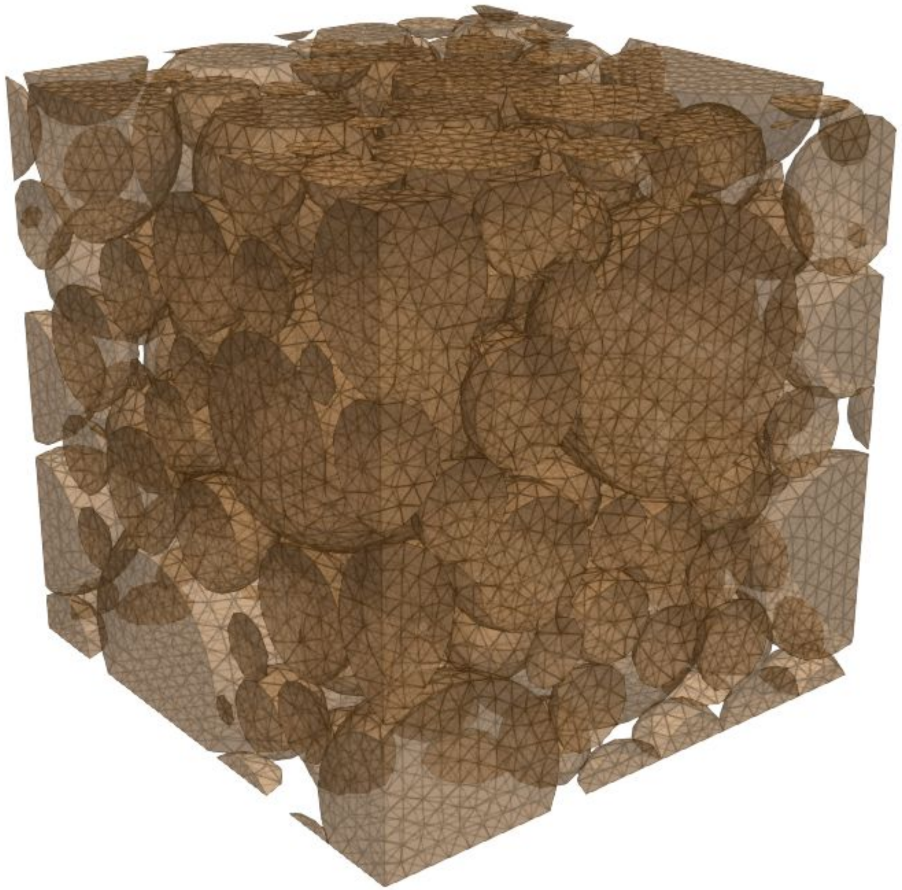
Pore scale fluid-structure interaction model

$p_{top} = 10\,kPa$

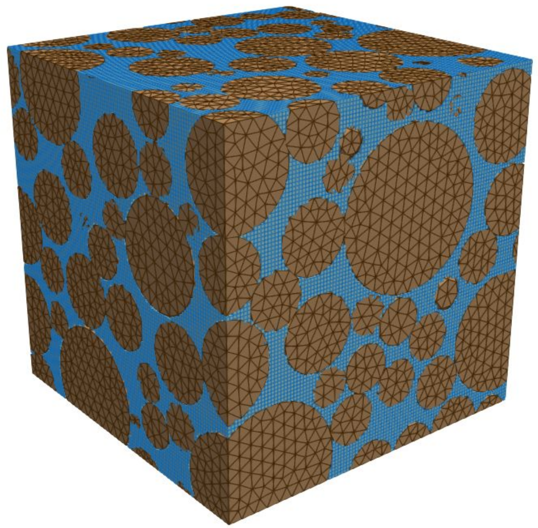


$p_{bot} = 0\,kPa$

Pore space grid



Grains grid



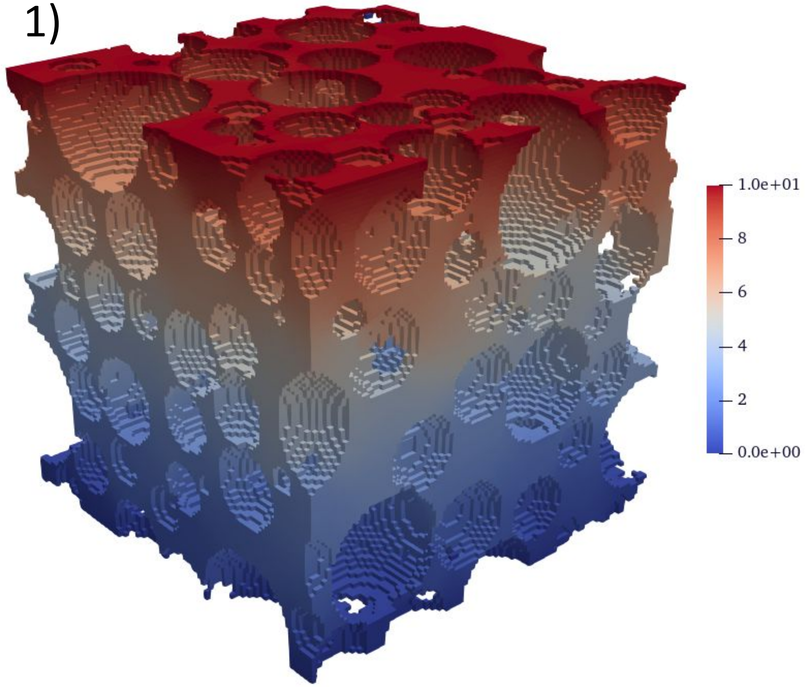
Pore space: 339468 cells
Grains: 19970 nodes
239 grains
140 mortar interfaces

Pore scale fluid-structure interaction model - results

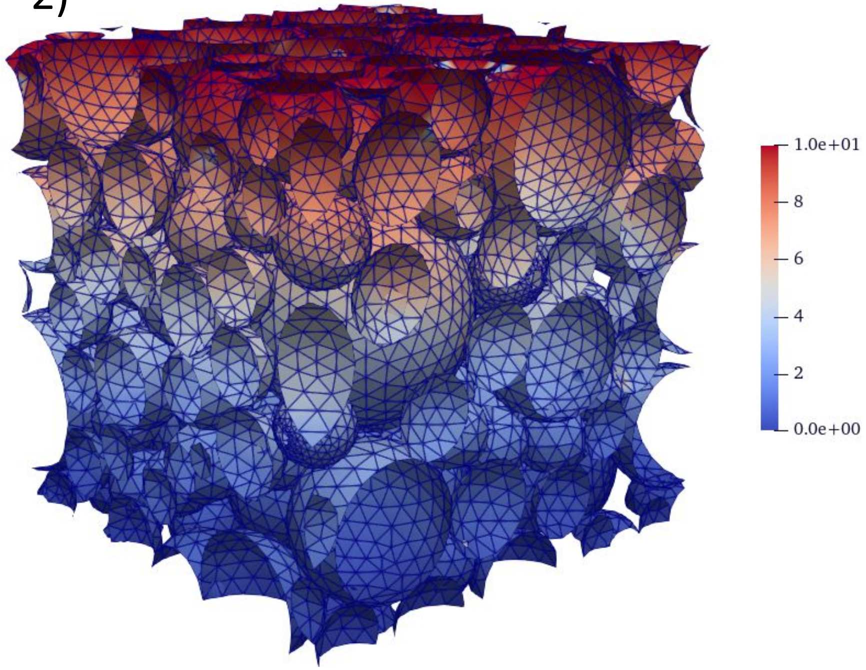
1. A finite volumes variably saturated flow simulation is run on the pore space grid
2. A staggered mortar operator is used to transfer the cell pressures to the grain surfaces
3. A mechanical simulation is run on the grains: tied contact between grains is simulated using fully coupled mortar simulation.

Flow simulation → Pressure field interpolated on the grain surface → **Mechanical simulation** displacement norm

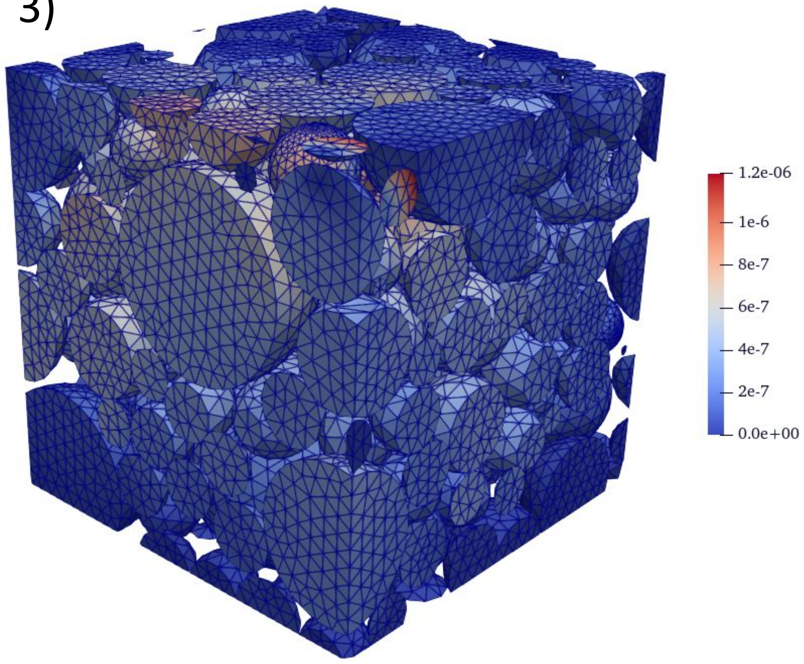
1)



2)



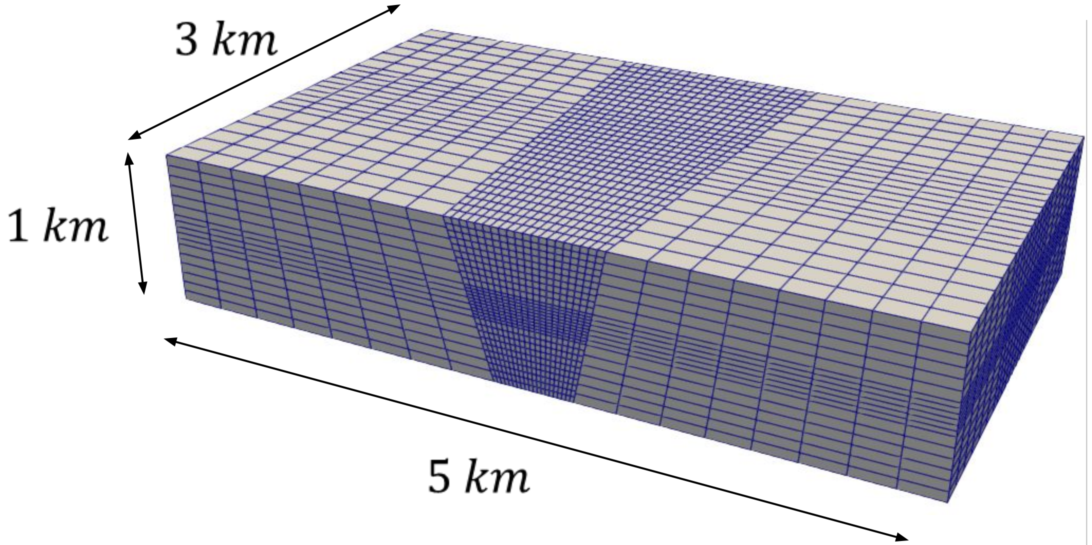
3)

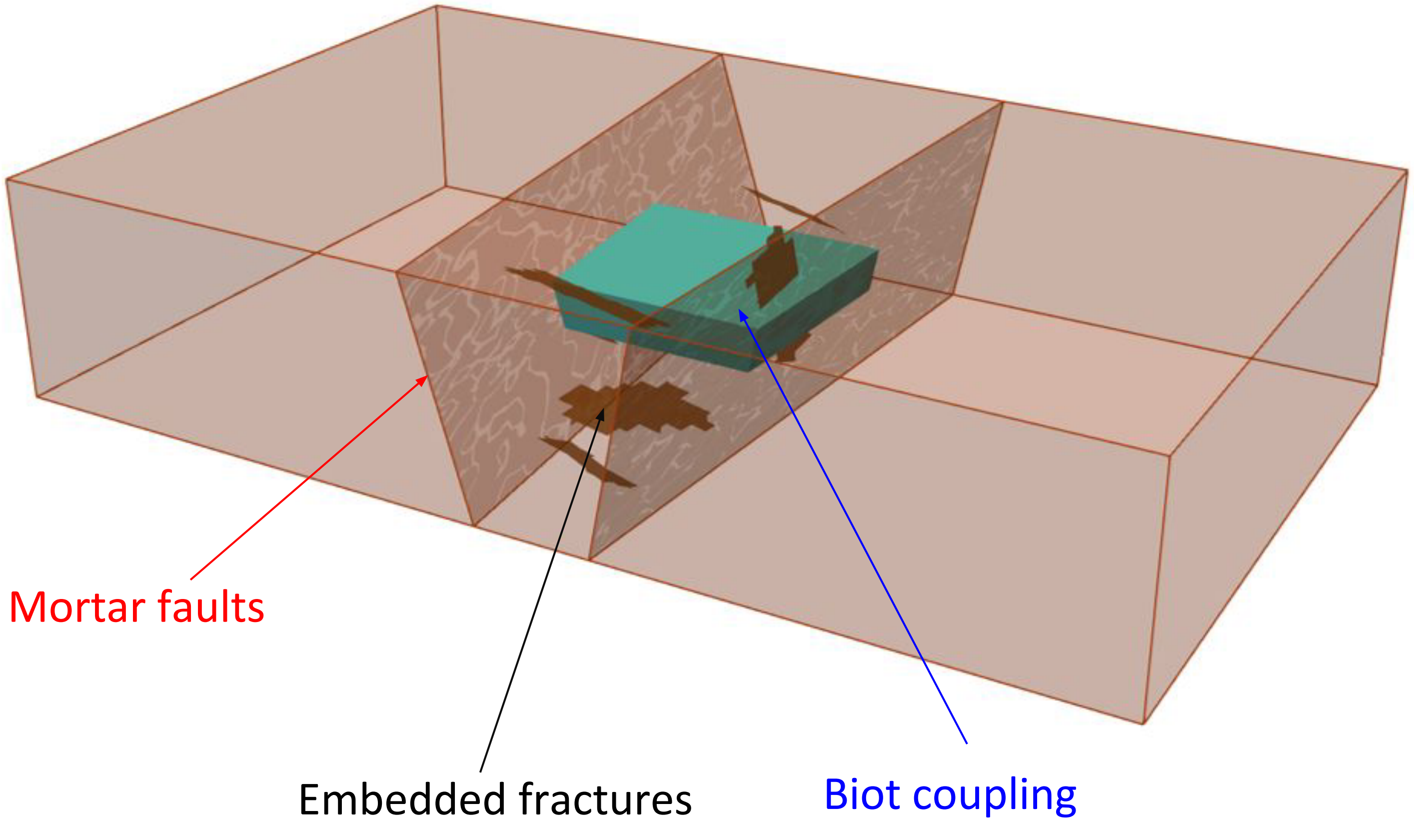


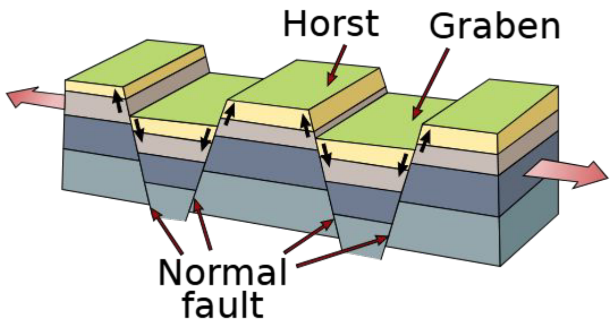
Results show that GReS is capable of handling any number of interfaces with significant geometrical complexity.

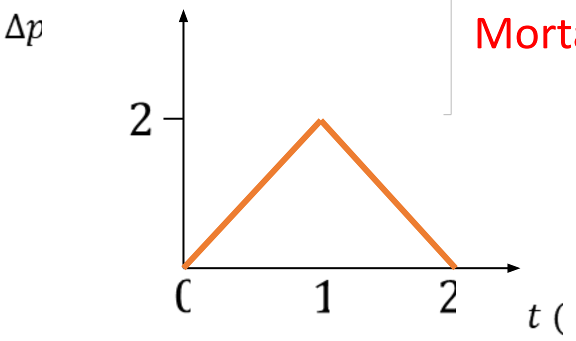
A realistic test case with mixed fracture representations

We simulate fluid extraction from a deep formation, bounded by two main fault planes. Several embedded fractures are introduced in the model.









Two years extraction program

Mixed fracture representation

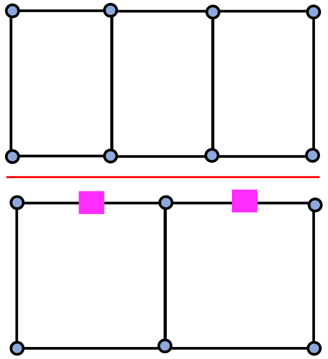
- The faults may be many, with varying size and geometrical complexities.
- Incorporating all the geological discontinuities in a single discretized model is very challenging.

We are interested in a unique modeling framework where regional and local faults are represented in different ways.

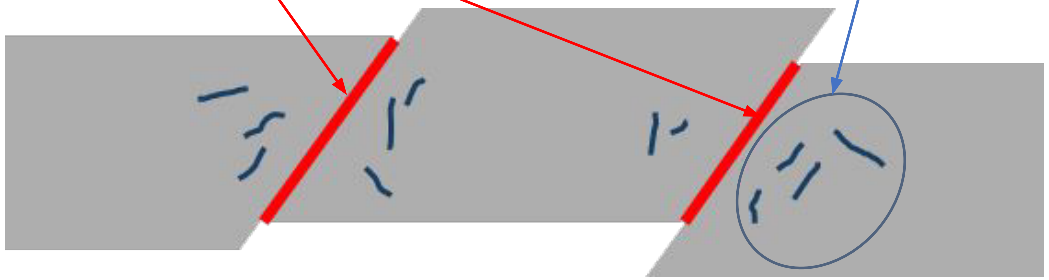
Major faults

- Larger size
- Few for each model
- Geometry is known

A traction field is defined in a lower dimensional interface altering the background grid



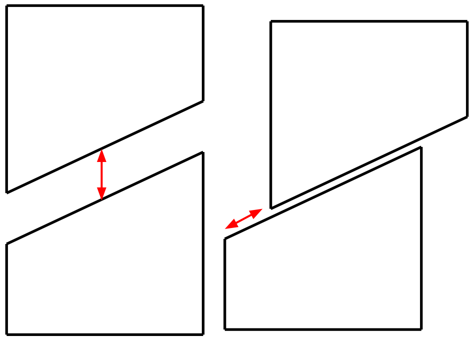
Displacement DoFs
Traction DoFs



Minor faults

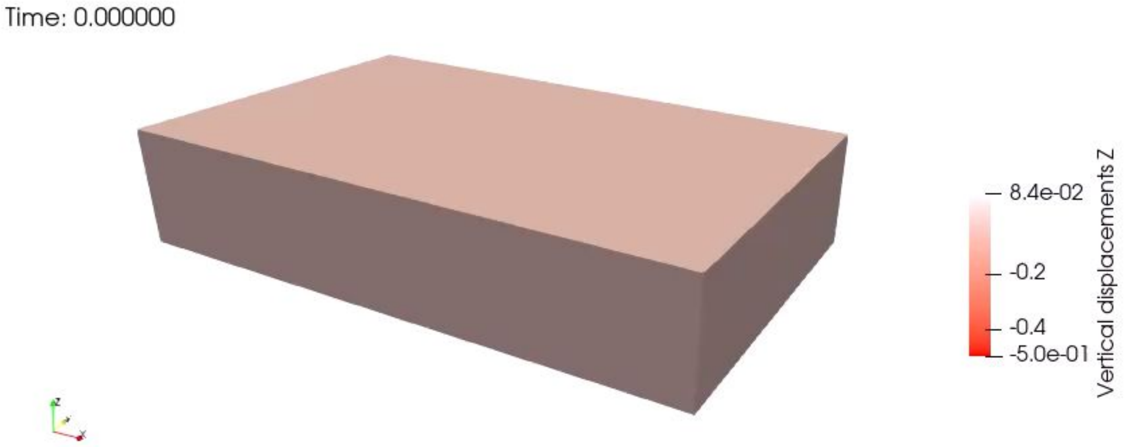
- Smaller size
- Many for each model
- Limited geometrical information

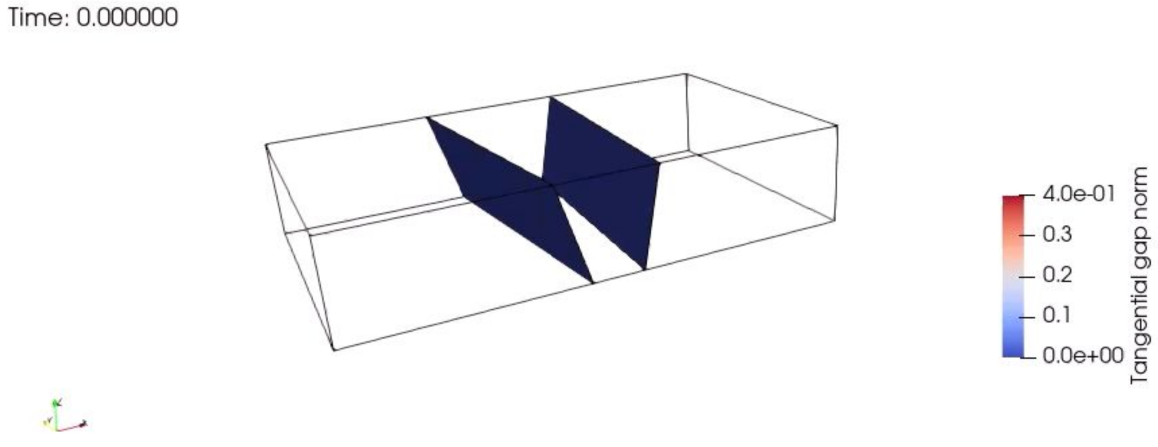
Fault are represented as discontinuity planes embedded in the background grid

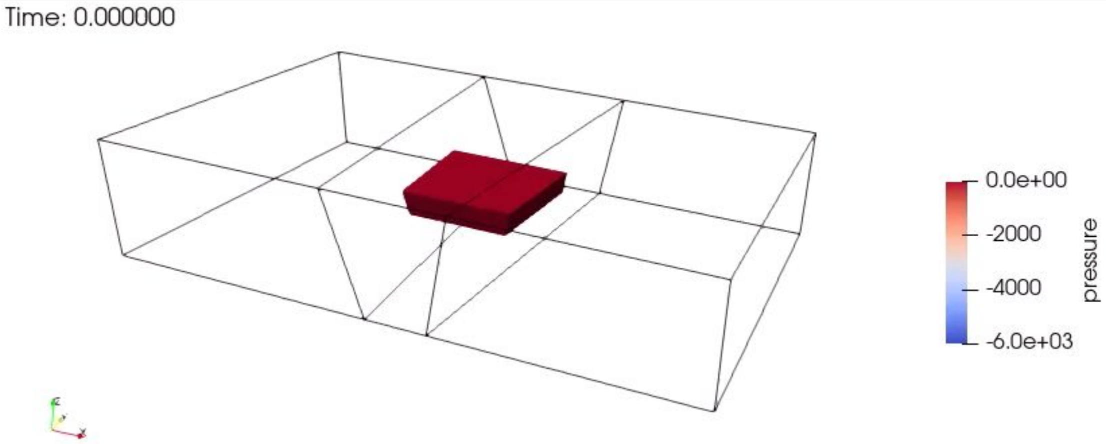


EFEM with P0 enrichment

All the processes are simulated simultaneously in a GReS fully coupled simulation







Conclusions

- A novel open-source MATLAB modular platform for coupled subsurface problems has been presented.

- The main focus is on multi-physics and multi-domain problems. The platform can simulate different physical processes on unrelated non-conforming grids.

- Open-source policy and high-level language: almost no effort to start using the code and anyone can contribute to it.

- The code has been used to prototype and test new fracture mechanics modeling frameworks combining different fracture representations.

Future developments

- Introduce additional constitutive laws for geomaterials;
- Introduce additional physical modules: heat transfer, multi-phase flow etc.;
- Implement preconditioning techniques for multi-domain coupled problems;
- Introduce non-conforming capabilities for Finite Volumes.
- Investigate robustness of contact algorithms on more challenging test cases.

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Thank you for your attention.